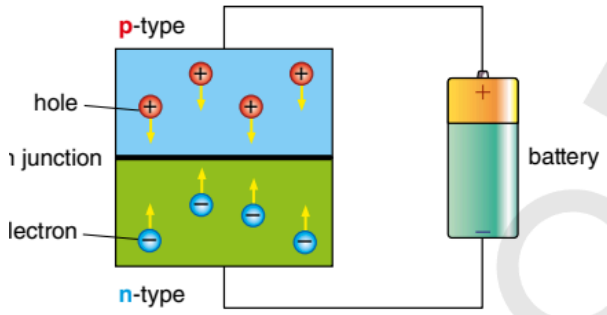
1. How LEDs produce light

Under forward d.c. bias, electrons and holes move towards the p-n junction and recombine. The energy decrease is released as electromagnetic radiation. The junction material and energy difference determine the emitted colour.



.1q An LED connected to a battery.

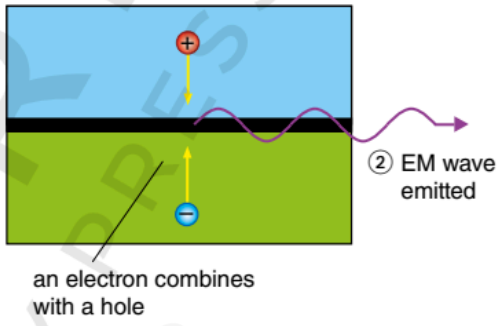


***Fig 2.1r** An electron combines with a hole and releases the excess energy as an EM wave*

Forward bias makes electrons and holes recombine at the junction and emit light.

2. Luminous flux

Luminous flux weights visible radiant output by the eye's wavelength sensitivity and is measured in lumens. Equal radiant powers with different spectra can have different luminous fluxes. Flux describes total visually effective light output or passage through a surface. Illuminance additionally divides received flux by area and has different units.

3. Illuminance

Illuminance is luminous flux received per unit surface area, measured in lux, equivalent to lm m^{-2}. Spreading the same flux over a larger area reduces average illuminance. The area-based calculation gives an average. With non-uniform lighting it does not imply the same illuminance at every point.

$$E = \frac{\Phi}{A}$$

E: illuminance (lx); Φ: luminous flux received by the surface (lm); A: illuminated area (m^2). Uniform illumination, or average illuminance over A.

4. Distance, incidence angle and illuminance

Light from an isotropic point source spreads out, so normal-incidence illuminance follows an inverse-square distance law. Oblique incidence spreads light over a larger surface, introducing the cosine of the angle from the normal. An angle measured from the surface needs conversion. Total source flux is not all received by a small target.

$$E = E_{\perp}\cos\theta = \frac{\Phi\cos\theta}{4\pi r^2}$$

Isotropic point source: Φ: total emitted luminous flux (lm); r: source-to-surface distance (m); θ: angle between incident light and surface normal; E: lx.

5. Luminous efficacy

Luminous efficacy is useful luminous flux per watt of electrical input, measured in lm W^{-1}. Higher values generally mean less electrical power for the same lighting requirement. It is not a percentage energy efficiency and its numerical value can exceed 1. Compare lamps using their flux and lighting conditions as well as efficacy.

$$\eta_v = \frac{\Phi}{P_{\text{in}}}$$

Luminous efficacy η_{v} (lm W^{-1}); Φ: useful luminous flux (lm); P_{in}: input electrical power (W). Not a dimensionless energy efficiency.